DRUMPANEL · CALCULATION

DrumPanel sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade and STEM fixes F1 to F4 (DMP-DEC-001): per-drum flatness check (section 5.3) and two drain stands (section 8), throughput 2.20 drums an hour; slot tolerance; cost USD 3,566; the script reads the station model (station_components()) on which the D3 model is built, so the geometry numbers are unchanged; section 5.1 reworded (a rolled hoop would buckle the panel; the steel would not tear), found by the STEM check (DMP-STM-001); requirement IDs now PR-001 to PR-012 (DMP-PRS-001), R numbers kept in brackets
0.2	2026-10-03	Amish Chadha	Amish's requirement decisions of 2026-10-03 (DMP-DDR-003); lever notching punch (new section 9), ring head and raised shear head drive, fine-pitch dial adjuster, throughput against R7 restated to 2.5 drums an hour, masses and cost
0.1	2026-10-03	Amish Chadha	

	First issue at TRL 3, on the constructable design of DMP-DDR-002

DrumPanel sizing calculations

On paper the DrumPanel bench set meets ten of its twelve requirements and misses one, and its estimated cost is over the value-engineering target. Every crank and lever force is at or below 136 N against the 150 N limit of R6, and the guards meet ISO 13857 on paper. R7, restated by Amish on 2026-10-03 to 2.5 drums an hour, is not met: with the lever notching punch, the ring head and the dial-set bending roll, two people would turn about 2.20 drums an hour into flat panels, up from 2.0, with two drain stands so the purge keeps up. R5 is met on paper with a trial pass for each batch and a straightedge check of the first panel of every drum: the slip roll leaves a panel flat only when its bending roll is within about 0.29 mm of the right height, a change of 0.02 mm of wall or 5 MPa of yield uses that up, and the fine-pitch dial sets and holds the height to 0.025 mm, about a twelfth of the allowance; the last 90 mm at each panel end is still set by hand. Value-engineering target: USD 3,000. Estimated cost of the constructable design: USD 3,566 (USD 566 over the target). Rolling the hoops flat would stretch their crowns about 8.5 %, more than three rolls can put in evenly, which is why the hoops are cut out and the body becomes three panels about 233 x 1,800 mm. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [D15], is the line of that script's output that carries it.

> **Safety:** These are first-principles estimates for a paper proof of concept. They do not show that a purged drum is free of vapour, that the guards are safe as built, or that the frames carry their loads in service. The vapour check, the guard openings and the drain stand must be checked on hardware before any drum is cut. See DMP-PRC-001, Safety.

Scope and method

The note checks every requirement of DMP-PRS-001 v0.6 (PR-001 to PR-012, which were R1 to R12 of DMP-REQ-001; the R numbers are kept below) against the design in DMP-PRC-001 v0.4 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, `levels()` and its station model `station_components()`, one solid per component as designed; the kit 1.9 detailed model of option A (`components("A")`, 818 parts with every bolt, weld and bearing, in the STEP files and drawing DMP-DWG-001) is built on the same parameters and positions, so the drum, rollers, shear head, slip roll and guards used here are the ones drawn. The detailed model, with both drain stands, weighs 403 kg against 417 kg for the station model here with two stands (hollow sections with real corner radii, holes and edge breaks; the drain trays in HDPE); the masses below stay on the station model. It reads `bom/bom.csv` for the cost and `budget_usd` in `project.yaml` for the target. Run it from the repo root with `python docs/04-calcs/sizing.py`; it also writes `docs/04-calcs/results.csv`. The model's own checks (`python cad/src/model.py --check`) show that no two of the station model's 52 components overlap, that every joint face touches, that the shear head with the ring head bolted on clears the drum in all three ring passes and alone at seven stations along the slit, that its crank sweeps clear of the drum, the posts, the rail beam and the trolley in both modes, and that the notching punch fits each of the four notches with its die faces on the steel.

Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Drum	572 mm inside diameter, 880 mm over the chimes, body 1.0 mm (0.8 to 1.2), heads 1.2 mm (1.0 to 1.5), hoops 298.5 mm outside radius at 147 mm each side of the middle	Common 55 US gallon tight-head drums
Steel	Yield 250 MPa (200 to 300), tensile strength 360 MPa, shear strength 0.8 of tensile, modulus 200 GPa, elastic and perfectly plastic in bending	Low-carbon drum sheet; screening values
Shearing	Force = shear strength x t^2 / (2 tan of the nip angle); feed resistance = force x tan of half the nip angle; discs overlap 1.0 mm	Standard rotary shear estimate
Edge spreading	Pushing the cut edges round the 10 mm web adds 100 N at 1.2 mm, scaling with the cube of the wall	Estimate, to confirm at TRL 4
Traction	Driven shear disc: effective friction 0.3 (the edge bites); knurled end cutter wheel 0.3; steel pinch rolls on painted steel 0.15	Screening values
Losses	20 % for the shear head bearings and idle disc; 30 % for the end cutter's rollers; bronze bushes friction 0.10; chain 95 %	Screening values
Slip roll	Panels fed three side by side (700 mm); pinch rolls 60 mm, bending roll 90 mm behind the pinch line; pinch force 2 kN set by two M16 screws	Model geometry
Notching punch	Inclined-edge shear per edge 0.5 x shear strength x t^2 / tan(rake), punch faces raked 10 degrees; the chime is five layers; 20 % for stripping; Tr24 x 5 screw, friction 0.15	Screening values; the chime force is to confirm on a scrap chime at TRL 4
Work rate	30 crank turns a minute; times for loading, setting up and handling as listed in section 8	Estimates for a practised two-person team
Purge	Hose at 20 L/min; 50 mm bung, discharge coefficient 0.6	Screening values

1. What a drum yields

After the two heads, the two chime rings and the two hoop strips are cut out, the body gives three panels and the heads give two discs.

Table 2. Output per drum.

Quantity	Value	Tag
Panel length (body circumference at mid-wall)	1,800 mm	[A1]
Panel widths, outer, middle, outer	233, 234 and 233 mm	[A2]
Head discs, two	552 mm across	[A3]
Flat area: panels; heads	1.26 m^2; 0.48 m^2	[A4, A5]
Empty drum; three panels at 1.0 mm	19.2 kg; 9.9 kg	[A6, A7]

The rest of the steel comes off as two chime rings and two hoop strips, about 4.8 kg, which artisans can use as strip.

2. Drain and purge

A full drum is filled and drained on its stand, never lifted. The stand slopes 2.9 degrees toward the bung end so it drains to within about 1.4 L.

Table 3. Purge and the drain stand.

Quantity	Value	Tag
Inside volume (brimful)	221 L	[B1]
Drum full of water	240 kg	[B2]
Fill time at 20 L/min	11.0 min	[B3]
Drain time through the 2 in bung	2.3 min	[B4]
Water left below the bung opening	1.4 L	[B5]
Contact on the chocks, from vertical	31.5 deg	[B6]
Normal force on each chock, full drum	691 N	[B7]
Bending stress in a saddle, full drum	10.4 MPa	[B8]
Compressive stress in a leg, full drum	1.3 MPa	[B9]

The stresses are small against the 275 MPa yield of S275; the stand is sized by handling and welding, not strength. The purge itself is a procedure, not a calculation: fill, soak 10 minutes with detergent, drain, then check (DMP-DDR-001, D1 and D2). Whether one fill is enough for a given drum is what the vapour check decides; a failed check means a second fill.

3. End cutter

The end cutter needs little force: the cut is short-stroke shearing of the head against the chime's inside wall.

Table 4. End cutter, 50 mm cutter wheel, 40 mm drive wheel, 150 mm crank.

Head thickness	Cut force	Feed resistance	Drive torque	Crank force	Tag
1.0 mm	429 N	70 N	1.8 N·m	12 N	[C10]
1.2 mm	571 N	100 N	2.6 N·m	17 N	[C12]
1.5 mm	807 N	156 N	4.1 N·m	27 N	[C15]

The drive wheel and guide roller must pinch the chime with about 520 N for the knurl to drive a 1.5 mm head [C20]; the cutter's clamp screw sets this. One trip round a head is 13.8 crank turns, about half a minute [C21, C22].

4. Rotary shear head

The shear head with the ring head bolted on is the hardest crank in the set: two ring cuts at once on a 1.5 mm wall need 136 N, inside the 150 N limit of R6.

The crank is no longer on the upper shaft. A 200 mm crank there swept through the drum wall in both slit and ring modes (the model's sweep check found about 13,700 mm³ of overlap), so it could not make a full turn. The crank now sits on a raised crank shaft in a drive case on the side of the head: a 12 tooth to 28 tooth #35 chain to a jackshaft, then a 15 tooth to 15 tooth chain down to the upper shaft, 2.33 to 1 overall, with a 155 mm crank (DMP-DDR-003, Q3). The reduction keeps two ring cuts on one crank within R6; the drive case is the chain guard.

Table 5. Shear head, 101 mm discs, 1.0 mm overlap, 155 mm crank through a 2.33 to 1 drive (chains 95 % each).

Wall	Separating force	Nip angle	Feed resistance	Crank force, one cut (slit)	Crank force, two ring cuts	Traction over resistance	Tag
0.8 mm	482 N	10.8 deg	85 N	16 N	32 N	1.69	[D8, D8r]
1.0 mm	713 N	11.4 deg	139 N	26 N	52 N	1.54	[D10, D10r]
1.2 mm	977 N	12.0 deg	213 N	39 N	79 N	1.38	[D12, D12r]
1.5 mm	1,429 N	12.8 deg	365 N	68 N	136 N	1.17	[D15, D15r]

Only the upper disc of each head is driven, because a gear between the discs would have to pass through the drum wall. With the ring head on, a coupling shaft joins the two upper shafts, so each head drives its own cut and the traction margin is the same as for one head. At 1.5 mm the driven disc has 17 % more traction than it needs; if it slips on a thick drum, the operator pushes the trolley along by hand while cranking. The separating force is carried round the frame: the 10 mm web sees 12.1 MPa and a 25 mm shaft 17.7 MPa [D20, D21]. At 30 crank turns a minute the discs now cut at 4.08 m/min [D22], about 2.3 times slower than a direct crank; the ring head more than wins this back by halving the number of ring passes (section 8). The trolley, drop bar and head with its drive weigh 28.2 kg and bend the rail beam 0.15 mm at mid-span [D23, D24].

4.1 Ring head

The ring head is a second shear head, made to the same drawings without a drive, that is lowered through the slit and bolted to the main head with a 10 mm spacer plate once the main head is swivelled for ring cuts (DMP-DDR-003, Q2). Its nip sits 233 mm along the drum from the main head's (234 mm for the middle pass), so the six ring cuts are made in three passes: the left chime ring with the left hoop's outer side, the two inner hoop sides, and the right hoop's outer side with the right chime ring. It weighs 20.4 kg with its spacer plate and coupling [D25]; with both heads at mid-span the rail beam bends 0.26 mm [D26], and the spacer plate carrying the ring head 233 mm out sees 24.4 MPa [D27].

5. Slip roll: reverse bending to flat

5.1 Why the hoops are cut out

A rolling hoop stands about 12 mm out from the body wall. Bent to the reverse curvature the slip roll needs, its crown would be stretched about 8.5 % [E29], more than ten times the 0.35 % surface strain of a plain panel in the rolls. Low-carbon drum steel would not tear at that strain, but three rolls cannot put it in evenly, so a rolled hoop would leave the panel buckled. So the hoop strips are cut out with the shear head (DMP-DDR-002, P7) and only flat panels go through the rolls.

5.2 The setting that leaves a panel flat

A panel leaves the drum curled to a 286.5 mm radius [E1]. To come out flat, it must be bent the other way past the elastic limit by just the right amount, so that it springs back to flat. For an elastic and perfectly plastic sheet the change of curvature x (in units of the elastic limit curvature) must satisfy x - M(x) / M_e = drum curvature / elastic limit curvature, where M / M_e = 1.5 (1 - 1 / (3 x²)). The script solves this for each steel and wall, then finds the bending roll height that bends the panel to that radius between the pinch line and the bending roll 90 mm behind it.

Table 6. Settings and forces for the three panels fed together (700 mm).

Yield	Wall	Reverse radius	Bending roll above lower roll	Moment	Bending roll force	Feed resistance	Tag
200 MPa	0.8 mm	278 mm	13.4 mm	21.5 N·m	239 N	71 N	[E200-8]
200 MPa	1.0 mm	345 mm	11.0 mm	33.9 N·m	376 N	112 N	[E200-10]
200 MPa	1.2 mm	411 mm	9.3 mm	49.1 N·m	545 N	163 N	[E200-12]
250 MPa	0.8 mm	225 mm	16.4 mm	26.6 N·m	295 N	87 N	[E250-8]
250 MPa	1.0 mm	278 mm	13.4 mm	41.9 N·m	466 N	138 N	[E250-10]
250 MPa	1.2 mm	331 mm	11.4 mm	60.8 N·m	676 N	201 N	[E250-12]
300 MPa	0.8 mm	189 mm	19.3 mm	31.6 N·m	351 N	103 N	[E300-8]
300 MPa	1.0 mm	234 mm	15.8 mm	49.9 N·m	554 N	164 N	[E300-10]
300 MPa	1.2 mm	278 mm	13.4 mm	72.5 N·m	805 N	239 N	[E300-12]

The nominal setting is 13.4 mm [E10]; the range over likely drum steel is 9.3 to 19.3 mm [E11], well inside the 60 mm of slot travel. The heights assume the panel follows a circular arc between the pinch line and the bending roll; with the moment falling to zero at the bending roll, as the force balance here has it, the independent review of DMP-STM-001 estimated about 9.7 mm nominal and 6 to 13 mm over the range. Either way the setting lies well inside the slot travel, and the trial pass, not the formula, finds it; the forces and moments are unchanged.

5.3 The trial pass and the dial adjuster (R5)

The setting is sensitive. If the nominal setting were used on steel at the edges of the range, a panel could sag up to 188 mm over 1 m [E12]. To stay within the 10 mm over 1 m of R5, the bending roll must be within 0.29 mm of the right height [E13]. With the former M20 x 2.5 screws and 12-division handwheels that allowance was only about 1.4 divisions [E14c], so the setting was at risk.

Each bending screw is now a fine-pitch M20 x 1.5 screw with an 80 mm dial of 60 divisions, 0.025 mm each, read against a pointer on the bridge, and a lock nut that holds the setting (DMP-DDR-003, Q4). The 0.29 mm allowance is 71 degrees of a turn, or 11.8 divisions [E14, E14d], so both ends can be set to the same reading well inside it and returned to after a change. The right height depends on the steel of each drum, not only of each batch: with the setting found for a 250 MPa, 1.0 mm drum, a wall 0.02 mm thicker or a yield 5 MPa higher already leaves about 8 mm of sag over 1 m, and 0.05 mm of wall or 10 MPa of yield about 16 to 20 mm (the independent review of DMP-STM-001, rechecked with this script's residual curvature). A reading found once for a batch of about ten drums, as v0.2 of this note had it, does not hold from drum to drum. So the first panel of each batch is passed, checked with a straightedge and the dials adjusted until it comes out flat; then on every drum the first panel is checked with the straightedge and the dials trimmed, and a panel that comes out of tolerance is passed again, turned over if it is over-bent (DMP-DEC-001, STEM fix F1). R5 is met on paper with these checks; a TRL 4 trial will show how many second passes they take. The bending roll itself bends about 0.07 mm under its worst load, a quarter of the allowance (DMP-STM-001 S26).

5.4 Panel ends

The slip roll cannot bend the leading 90 mm of a panel (from the pinch line to the bending roll) [E15]; left alone, that end would stand 14.1 mm above flat [E16]. The ends are set flat by hand with a dead-blow mallet over the edge of the bench plate. This is quieter and far shorter work than the hammering it replaces.

5.5 Drive, rolls and bushes

Table 7. Slip roll drive and parts, worst case (300 MPa, 1.2 mm).

Quantity	Value	Tag
Bending roll force	805 N	[E17]
Feed resistance	239 N	[E18]
Pinch force needed (friction 0.15, margin 1.5)	1,193 N	[E19]
Pinch force set by the two M16 screws	2,000 N	[E20]
M16 screw torque for 1 kN; force on the 90 mm handwheel rim	3.2 N·m; 71 N	[E21]
Torque to turn the pinch rolls	16.8 N·m	[E22]
Crank force at 300 mm through the 3:1 chain	20 N	[E23]
Feed speed at 30 crank turns a minute	1.9 m/min	[E24]
Upper pinch roll load; mid-span deflection	2,805 N; 0.25 mm	[E25]
Upper pinch roll bending stress	15.5 MPa	[E26]
Bronze bush pressure	2.3 MPa	[E27]
Pinch gear tooth stress (Lewis)	15 MPa	[E28]

Every part runs far below its limits; the rolls are 60 mm because they must stay straight to a fraction of the 0.29 mm setting tolerance, not for strength.

6. Cutting cradle

The drum sits on its hoops at 33 degrees from vertical each side [F2], with its axis 603 mm and its top 890 mm from the floor [F1], a comfortable height for the crank work. With 1 kN on the drum (the drum plus a person leaning on it) a roller shaft sees 72 MPa and bends 3.2 mm at mid-span [F3]; under the drum alone it bends about 0.6 mm. A sideways push of 122 N rolls an empty drum over a roller [F4], so the drum brake is used whenever the drum must stay still.

7. Guards (ISO 13857, slot openings)

ISO 13857 Table 4 allows a slot opening of up to 8 mm when the danger point is at least 20 mm behind it; a slot over 8 mm needs 80 mm, so the slots are drawn at 8 mm and made 7 to 8 mm (PR-010, STEM fix F4).

Table 8. Guard openings.

Opening	Slot	Distance to the nip	Needed	Result	Tag
Slip roll in-feed slot under the nip guard	8.0 mm	45 mm	20 mm	Meets	[G1]
Slip roll out-feed slot	8.0 mm	85 mm	20 mm	Meets	[G2]
Shear head front skirt over the sheet	7.5 mm	52 mm	20 mm	Meets	[G3]

The slip roll chain, sprockets and pinch gears are fully enclosed, and so are the shear head's two chains, inside its drive case. The coupling shaft between the main head and the ring head is a smooth 25 mm shaft turning at about 13 revolutions a minute inside a loose guard sleeve. The end cutter's wheel works inside the head recess under a sheet cover. The notching punch moves only as its screw is turned by the lever, with the hands 480 mm from the punch. These are desk checks on the model; a competent person checks the guards as built before any trial.

8. Throughput

Two people turn about 2.20 drums an hour into flat panels [H5], against the 2.5 of R7 as restated by Amish on 2026-10-03. R7 is not met: 2.5 drums an hour needs 48.0 hands-on minutes per drum [H9], and the estimate is 54.6 [H4].

Table 9. Hands-on minutes per drum.

Station	Before (DMP-CAL-001 v0.1)	Now	Tag
Purge and vapour check	14	14	[H1]
Cutting cradle: load, two heads, four notches, slit, six ring cuts, unload	30.1	23.6	[H2]
Slip roll and finishing: setting and per-drum check, three passes, handling, panel ends, deburring	17.1	17.1	[H3]
Total	61.2	54.6	[H4]
Drums per hour for two people	2.0	2.20	[H5]

What changed at the cradle [H8]: the four notches take 4.0 minutes with the lever notching punch, down from 8 with a hacksaw (1.2 minutes for each chime notch, 0.8 for each hoop slot [K9]). The six ring cuts take 9.5 minutes in three passes with the ring head, down from about 12.4 in six, including a minute a drum to bolt the ring head on and take it off. The slit takes 2.7 minutes, up from 2.3, because the new drive turns the discs 2.33 times slower. At the slip roll the trial pass is taken as 10 minutes for a batch of about ten drums (1.0 minute a drum), then 1.5 minutes a drum to check the first panel with the straightedge and trim the dials, and 0.5 minutes a drum for second passes (about one panel in three), 3.0 minutes a drum in all (STEM fix F1; 1.5 minutes in v0.2). The cutting times use an effective speed of one third of the disc speed at 30 crank turns a minute, an estimate that allows for hand changes on the crank, stops to ease the cut edges round the web and the start of each pass.

The purge takes about 23.3 minutes elapsed and runs while the cradle works [H6], but it holds its stand for the whole cycle: loading, fill, soak, drain, tipping and the check take about 35.3 minutes a drum [H10], so one stand could purge only about 1.7 drums an hour [H11], less than the cradle and slip roll use. The independent review of DMP-STM-001 found this; the set now has two drain stands and trays (DMP-DEC-001, STEM fix F2), which purge about 3.4 drums an hour [H12], so the hands-on rate of 2.20 holds [H13]. A drum that fails the vapour check takes another cycle. Taking both heads off is about 3.9 minutes [H7]. The remaining gap of about 6.6 minutes a drum is spread over handling, setting up the ring passes and deburring; ways to close it are listed in the design decisions register (DMP-DEC-001, Open decisions and Value engineering). Even at 2.20 drums an hour, a two-person team turns a drum into panels in under 30 minutes, against the hours of fire and chisel work the method replaces.

9. Lever notching punch

The discs cannot start a cut through a folded chime or a hoop ridge, so four notches are made on the slit line first: an open-ended notch 60 x 35 mm at each drum end through the chime, and a slot 40 x 32 mm through each hoop. The lever notching punch replaces the hacksaw for these (DMP-DDR-003, Q1). It is a C-frame that slides into the open drum end with its lower jaw inside the drum: station 1, at the back of the frame, notches the chime with the frame's back face on the drum end; station 2, 338 mm out, slots the hoop with the drum end against the station 1 die block, its lower jaw passing in through the chime notch already cut. Each station has a D2 punch with its face raked 10 degrees, pushed into a matching die by a Tr24 x 5 screw turned by a 500 mm ratchet lever.

Table 10. Lever notching punch, 1.5 mm steel (the worst case of the accepted range).

Quantity	Chime notch	Hoop slot	Tag
Punch force	24.3 kN	6.6 kN	[K1, K2]
Screw torque; lever force at 480 mm	63 N·m; 131 N	17 N·m; 36 N	[K3, K4]
Jaw root bending stress, lower and upper jaw	18 and 18 MPa	96 and 93 MPa	[K5, K6]
Lever swings under load (60 degrees each)	18	8	[K8]
Time per notch	1.2 min	0.8 min	[K9]

The chime notch is the hard one: the punch shears five layers of the folded seam. The lever force of 131 N is within R6, and the jaws, S355 plate, are at about a third of yield at the hoop station. Under the hoop slot force the jaws open about 1.3 mm at the hoop station [K7]; the punch is guided in its boss, so this is taken up in the screw stroke rather than shifting the punch off its die. The chime force is the least certain number here, because a real chime seam's layers and its clinch vary; it is to be confirmed on a scrap chime at TRL 4. The punch weighs 18.7 kg and is carried to the drum by one person [I9].

10. Masses

Table 11. Masses (steel at 7,850 kg/m³; chocks, tray, bushes and plywood at their own densities).

Item	Mass	Tag
Drain and purge stand with tray, each of two	15 kg	[I1]
Cutting cradle with posts, beam, shear head, ring head, end cutter, notching punch and tool shelf	196 kg	[I2]
Slip roll stand with tables	191 kg	[I3]
Heaviest lift: cradle frame, one welded piece	33.7 kg	[I4]
Shear head with shafts, discs, guard, drive and drop bar	23.7 kg	[I5]
Lower pinch roll	21.0 kg	[I6]
Ring head with its spacer plate and coupling	20.4 kg	[I7]
Side frame with legs and bridge, each	20.1 kg	[I8]
Notching punch, carried to the drum	18.7 kg	[I9]

Only the cradle frame is over the 25 kg one-person limit of R11; it is lifted by two people. The shear head with its drive (23.7 kg) and the ring head (20.4 kg) are under the limit but awkward to hold while bolting; the build plan fits each with a second person steadying it.

11. Cost

Value-engineering target: USD 3,000. Estimated cost of the constructable design: USD 3,566 (USD 566 over the target) [J1 to J3]. The second drain stand and tray of STEM fix F2 add USD 77. The decisions of 2026-10-03 added USD 695: the notching punch (USD 330, replacing the USD 45 hacksaw line), the ring head (USD 255), the shear head drive (USD 75), the dial adjusters (USD 40) and the tool shelf (USD 40). The largest lines are now the vapour check kit (USD 600), the shear head (USD 340), the notching punch (USD 330) and the three rolls (USD 285) [J4 to J7].

12. Results against the requirements

Table 12. Requirement status at TRL 3.

ID	Requirement	Result	Status
R1	No open flame	All steps cold	Met by design
R2	Vapour check flags 10 % LEL	Detector alarm at 5 % LEL	Met by design
R3	Both heads off in under 10 min	3.9 min [H7]	Met on paper
R4	Straight slit, 5 mm	Rail-guided head	Met by design
R5	Flat within 10 mm over 1 m	Needs 0.29 mm setting accuracy [E13]; dial reads 0.025 mm, the allowance is 11.8 divisions [E14d]; trial pass for each batch, first panel of every drum checked and the dial trimmed; ends hand-set [E16]	Met on paper, with the per-drum check
R6	Crank force at or below 150 N	136 N worst, two ring cuts at 1.5 mm [D15r]; punch lever 131 N [K3]	Met on paper
R7	2.5 drums an hour, two people (restated 2026-10-03)	2.20 drums an hour [H5, H13], two drain stands	**Not met**
R8	Local build	Welding, drilling, turning; bought discs and bearings	Met by design
R9	Value-engineering target USD 3,000	USD 3,566 [J1]	Over the target by USD 566
R10	Guarding	All slots meet ISO 13857 on paper [G1 to G3]	Met on paper
R11	Handling	33.7 kg frame by two people [I4]; every other piece 23.7 kg or less [I5 to I9], the shear head and ring head lifted to the rail by two people; full drum never lifted	Met with a two-person rule
R12	Yield	Three panels 233 to 234 x 1,800 mm, two 552 mm heads [A1 to A3]	Met by design

Where the numbers come from

- `docs/04-calcs/sizing.py` (this note's script) and `docs/04-calcs/results.csv`.
- `cad/src/model.py`: geometry, `PARAMS`, `levels()`, fit checks.
- `bom/bom.csv` and `project.yaml`.
- ISO 13857:2019, Safety of machinery: safety distances to prevent hazard zones being reached by upper and lower limbs, Table 4.